SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

DenTrack: Comprehensive Cloud & Offline-First Dental Practice Management Platform

Version: 1.0.0
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Status: Final Specification
Target Architecture: Flutter, FastAPI, & Supabase Open-Source Ecosystem

1. Introduction

This Software Requirements Specification (SRS) document details the complete functional and non-functional requirements for DenTrack, an enterprise-grade, cloud-synchronized, and offline-first Dental Practice Management Platform. The system is designed to seamlessly serve clinical, administrative, financial, and patient-facing workflows within solo practices, multi-chair clinics, and distributed hospital network groups.

1.1 Purpose

The purpose of this document is to establish a rigorous, unambiguous baseline for the engineering, architecture, implementation, and quality assurance phases of DenTrack. It serves as the primary technical contract between developers, UI/UX designers, system architects, and healthcare stakeholders, ensuring all clinical compliance and workflow goals are precisely addressed.

1.2 Scope

DenTrack comprises a multi-tier cross-platform ecosystem optimized for cross-functional user roles. The scope spans an offline-first desktop clinical installation for operators requiring absolute zero-latency interface feedback, a cloud-backed receptionist and administration web portal, native iOS and Android patient engagement portals, and automated external integrations (WhatsApp/SMS gateways, laboratory tracking systems, and AI computer-vision pipelines).

1.3 Definitions, Acronyms, and Abbreviations

Acronym / Term	Definition / Description
EDR	Electronic Dental Record - Digital clinical charts detailing a patient's historical and contemporary dental status.
IOPA	Intraoral Periapical Radiograph - Detailed X-ray displaying the full tooth structure from crown to root structure.
OPG	Orthopantomagram - Panoramic scanning dental X-ray of the upper and lower jaw anatomy.
CBCT	Cone Beam Computed Tomography - 3D volumetric imaging technique capturing bone structures and soft tissues.
STL / OBJ	Standard Tessellation Language / Object File - Open 3D mesh formats mapping intraoral digital impressions.
RBAC	Role-Based Access Control - Security framework restricting application system permissions based on organizational rank.
HIPAA	Health Insurance Portability and Accountability Act - Regulatory framework defining data

	privacy and security provisions for safeguarding medical info.

2. Open-Source & Budget-Optimized Tech Stack

To deliver premium, real-time performance with zero initial overhead costs, the platform utilizes an optimized open-source, client-heavy engineering architecture. Workloads are shifted to client endpoints, minimizing server-side computational requirements.

Layer	Technology Selected	Architectural Rationale & Scale Advantages
Frontend Framework	Flutter (Desktop/Mobile)	Enables a shared native codebase for high-performance desktop applications (Windows/macOS) used in clinical ops and cross-platform native mobile patient applications.
Web Administration Portal	Next.js & React	Provides server-side rendered, search-optimized responsive web layouts for clinic setup, admin controls, and browser-based patient access.
Backend REST & WS API	FastAPI (Python)	High-performance asynchronous framework natively compatible with data-science, image-processing, and machine-learning dependencies (OpenCV, PyTorch).
Core Database & Infrastructure	Supabase (PostgreSQL)	Open-source Firebase alternative providing full relational schemas, Row-Level Security (RLS), instant JWT authentication, and automated real-time sockets.
Local Offline Storage	Drift (SQLite Engine)	Embedded relational cache matching the remote schema structure exactly to facilitate zero-latency offline processing and state caching.
3D Graphics Pipeline	Three.js / Flutter Mesh Viewers	Implements client-side client GPU rendering via WebGL to view raw .STL and .OBJ intraoral optical impression

	scans without specialized server resources.

3. Functional Requirements & Comprehensive Module Specifications

This section contains comprehensive structural breakdowns of the 24 foundational modules within DenTrack.

1. Patient Management

Supports patient onboarding, unique auto-incrementing ID token generation, multi-parameter demographic tracking, contact definitions, emergency nodes, granular medical/dental charts, allergy tags, dynamic lifestyle choices (smoking, alcohol metrics), electronic consent acquisition, and secure profile photography storage.

2. Electronic Dental Records (EDR)

Maintains chief complaints, history of present illness (HPI), structural clinical evaluation documentation templates, diagnostic indexing (ICD-10-CM / SNODENT standard references), multi-phase treatment frameworks, time-stamped progress logs, and encrypted digital signatures for procedural validation.

3. Interactive Odontogram

Features a dynamic vector graphics visual representation of permanent (32 teeth) and primary (20 teeth) dentition. Tracks existing conditions, caries, root-canals, extractions, implants, crown restorations, and fixed partial dentures with distinctive color overlays. Provides historical event tracking down to the individual tooth surface level.

4. X-ray & Diagnostic Media Workspace

Manages storage, version tracking, and visual display for IOPA, OPG, CBCT, cephalometric, and clinical photographs. Supports local, browser-based image modifications (brightness adjustments, histogram balance, linear measurements, markup layers) along an interactive chronological timeline. Handles raw .STL/3D formats via local GPU allocation.

5. Treatment Planning & Optimization

Enables production of alternative treatment options, detailed tooth-by-tooth sequence maps, automated pricing estimates, linked scheduling paths, explicit consent tracking, progress mapping, and step-by-step procedure check-offs.

6. Clinical Procedure Templates

Pre-built operational blueprints for Scaling, Composite Fillings, RCT, Crowns, Extractions, Implants, Orthodontics, and Maxillofacial options. Auto-populates notes, expected materials, standard billing line items, and estimated room duration metrics.

7. Multi-Chair Appointment Management

Provides real-time interactive calendar displays tracking doctor availability, chair utilization, waiting list updates, and automated notification hooks (WhatsApp/Email). Tracks cancellations and reschedules to manage clinical efficiency.

8. Automated Billing & Invoicing Engine

Generates detailed billing paperwork, itemized invoicing, split or partial payment flows, outstanding ledger analytics, localized tax (GST) modules, payment confirmations, and granular trackable adjustment configurations.

9. Treatment Packages & FinTech Plans

Combines multiple multi-visit procedures into comprehensive packages (e.g., Full Mouth Rehabilitation, Smile Makeovers) with automated payment installations, flexible pricing structures, and installment processing logs.

10. Digital Prescription Platform

Integrated medication lookup database with dosage calculators, systemic interaction cross-checks, customizable instructions, digital signatures, and automated PDF export systems.

11. Laboratory Work Order Ecosystem

Tracks technician workflows, order dispatches, digital shade matching logs, structural material specifications (Zirconia, PFM, Emax), delivery tracking status, and balance payments.

12. Inventory & Supply Chain Controller

Real-time stock management covering materials, bone materials, bonding agents, burs, and accessories. Features automatic barcode/QR capture scripts, expiration alerts, and minimum-quantity reorder flags.

13. Advanced Implant Tracking Module

Maintains comprehensive technical variables per patient including system brand, exact fixture sizing, factory batch details, healing configurations, surgical insertion torque (Ncm), measured bone density (Hounsfield units), and automated PDF implant passport documents.

14. Comprehensive Orthodontic Module

Tracks cephalometric analytic values, specialized multi-angle photography progress grids, aligner staging numbers, and custom tracking metrics for scheduled tracking checks.

15. Prosthodontic Workflow Module

Documents detailed preparation phases including jaw relationship registers, shade maps, custom tray configurations, try-in milestones, delivery validations, and warranty certificates.

16. Proactive Recalls & Follow-up Rules

Automated scheduler scanning parameters to generate SMS/WhatsApp notifications for routine cleanings, implant reviews, post-extraction assessments, and restorative maintenance windows.

17. Unified Patient Communications Engine

Omnichannel communication hub integrating WhatsApp Business APIs, SMS gateways, and SMTP servers to send system reports, confirmations, billing statements, and clinical material updates.

18. Analytical Reporting & Business Intelligence

Aggregates key metrics into accessible dashboards tracking daily/monthly collections, accounts receivable, new case growth, case conversion efficiencies, procedure revenue distributions, and individual provider production summaries.

19. Granular Role-Based Access Controls

Enforces strict user-group permissions across administration, clinical providers, front desk staff, surgical assistants, and accounting personnel to manage data access limits.

20. Multi-Region Cloud Sync & Security

Handles distributed database availability, automated backup schedules, system transaction logs, custom session controls, and strict multi-factor authentication requirements.

21. Artificial Intelligence Integration Frameworks

Optional processing layers for machine-learning tools, supporting automated 2D radiograph abnormality discovery, automated charting outlines, voice-directed dictation translation, and cost forecasting loops.

22. Dedicated Patient Portal Application

Accessible cross-platform application for patients to manage appointments, check treatment histories, view diagnostics, complete outstanding invoices, and view educational resources.

23. Comprehensive Document Repository

Central repository managing uploaded external files, insurance policies, clinical referral records, and scan results via structured access points.

24. Operational Clinic Cockpit Dashboard

Central control room layout displaying live clinic volumes, pending queues, collection details, processing lab alerts, inventory warnings, and tracking indicators.

4. Non-Functional Requirements

4.1 Security & Regulatory Compliance Compliance

• **Encryption Standards:** All communication paths require TLS 1.3 encryption layers. Storage volumes inside the Supabase cloud cluster deploy hardware-enforced AES-256 configurations. Field-level cryptographic structures protect sensitive items (e.g., systemic health profiles including HIV, Hepatitis markers, and high-risk medical alerts).
• **Database Row-Level Security (RLS):** PostgreSQL RLS policies are enforced on every table. Queries without valid user JSON Web Tokens (JWT) are rejected at the database engine layer, preventing cross-tenant data leakage.
• **Immutable Audit Trail:** Every record update, reading, or deletion creates a non-modifiable audit entry tracking timestamp details, operator IDs, IP addresses, and differences between old and new record values.

4.2 System Performance & Scalability Boundaries

• **UI Frame Rate Target:** The native Flutter clinical client UI must maintain a stable 60 Frames Per Second (FPS) target across regular clinical workstations, even when handling complex vector graphics charts or 10,000+ patient database searches.
• **Offline Synchronous Delay:** When recovering from a network drop, the client synchronization engine must reconcile local transaction stacks with the master cloud database within 5 seconds over standard 3G mobile connections.
• **Data Transmission Performance:** Standard REST API endpoints must return payload data within 200 milliseconds under normal usage conditions, excluding intensive 3D file or large image loading times.

4.3 Reliability & Availability Metrics

The core cloud database infrastructure target availability is 99.95% uptime, excluding pre-announced operational maintenance windows. The platform's offline-first architecture ensures that clinical workspaces remain fully functional for data entry, treatment logging, and scheduling updates during local ISP network interruptions, maintaining operational continuity.